

Say hello to GeoServer 3



GeoServer



Introductions

Jody Garnett

Product Owner

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[@jodygarnett](https://twitter.com/jodygarnett)



GeoCat

Netherlands open-source company focused on spatial data publication and discovery.



Open Source Projects

- GeoServer, GeoTools, JTS, ImageN, uDig

Open Source Geospatial Foundation

- Incubation Committee Chair
- GeoTools Project Officer
- GeoServer Project Steering Committee
- Marketing Committee

Eclipse Foundation

- JTS Topology Suite
- ImageN

Introductions

Ian Turton

PSC Member

ian@ianturton.com



Open Source Projects

- GeoServer, GeoTools, JTS

@ianturton@mapstodon.space

Ian Turton Ltd

Computational geographic consultancy

GeoServer at a Glance



Java Web Application to share and edit geospatial data.

Publish data from any major spatial data source using open standards.

Core Contributors



A F R I G I S

camp**to**camp

2026 Sponsors

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How 2 Map



Core Protocols

WMS – maps

WFS – vector

WFS-T – editing

WCS – coverage

WMTS – tiles

TMS – tiles

WMS-C – tiles

Extension/community protocols

WPS – process

CSW – search

OGC API - Features - vector

OGC API - json + rest standards

OGC STAC - spatio-temporal asset catalogue

Why GeoServer 3?

2024 Roadmap Challenge

Get ready for a ton of upgrades!

GeoServer has a large number of dependencies (upstream libraries)

Sometimes they decide to do breaking changes...

Sometimes they decide to make a lot of them... maybe multiple libs at the same time

This is one of those times!

And we need to follow, to keep up to date and **get security updates. But it snowballed... into an avalanche**



The Java upgrade avalanche.... started in

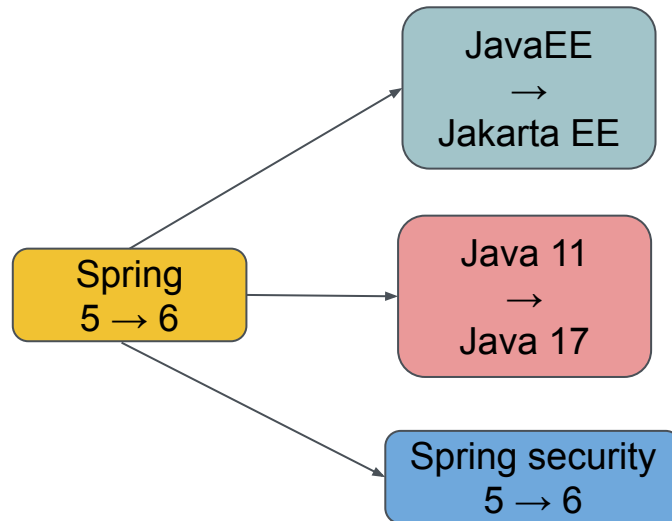


GeoServer is built with Spring, a popular framework used for web applications.

GeoServer 2 used Spring 5.3:
EOL **August 2024**
No more security updates.

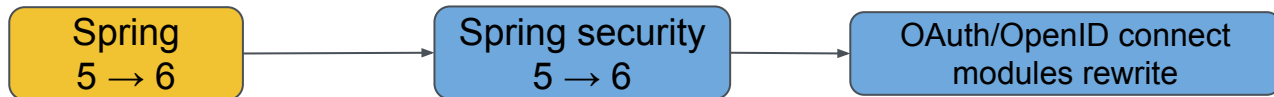
Spring 6 requires:

- Java EE $\rightarrow$ Jakarta EE
- Java 11 $\rightarrow$ Java 17



Spring-security update too

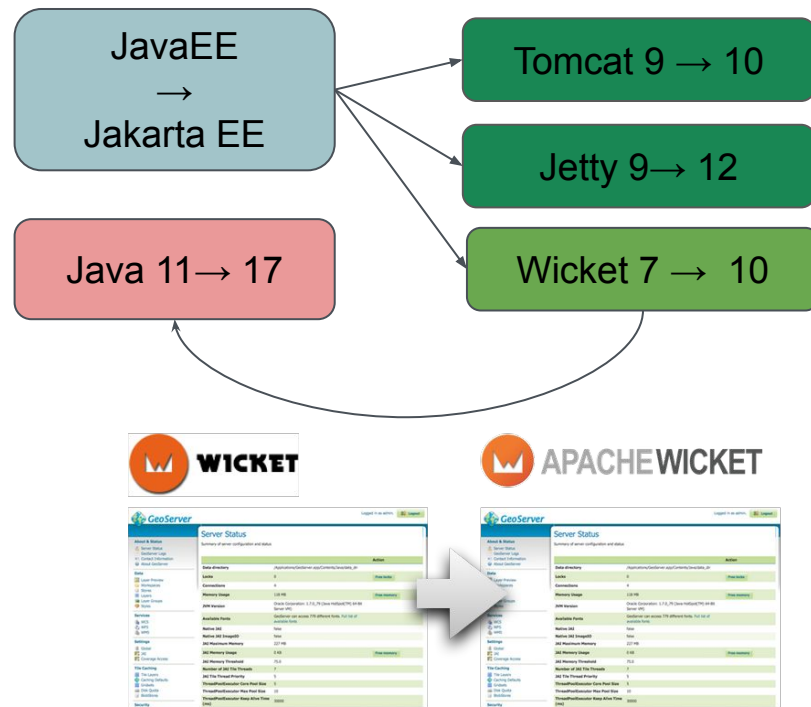
- We need to upgrade Spring Security to 6
- Incompatible changes
 - CAS support
 - OAuth support
- Popular community modules to rewrite!
 - OAuth
 - OpenID connect



Java EE to Jakarta EE

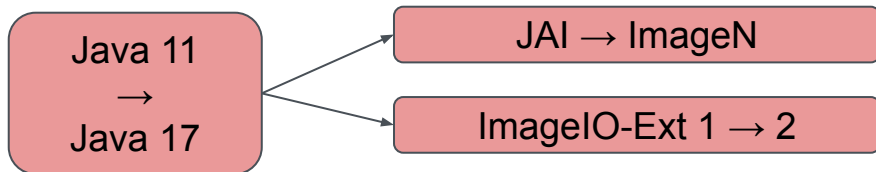
Java Enterprise Edition was donated to the Eclipse Foundation as the new **Jakarta EE** project (basic concepts for web application).

- New Tomcat and Jetty for Jakarta EE
- Wicket 7 uses Java EE, we need Wicket 10!
- and then Wicket 10 uses Java 17



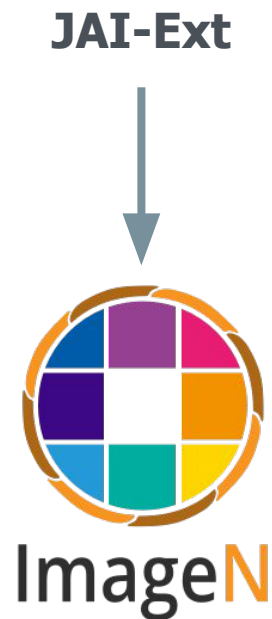
Java 11 to Java 17

- Spring 6 and Wicket 10 require **Java 17**
- We need to migrate the whole codebase to use **Java 17**, plus some dependencies
- Java 17 has new functionality, also strict!
- Java Advanced Imaging was part of Java **not donated to OpenJDK**
 - Used for raster processing, performs a few operations as Java 8 plugin, that are not allowed when used as a Java 17 library
 - Warnings for the moment, but eventually they will be blocked!
- **We need to do something about JAI!**

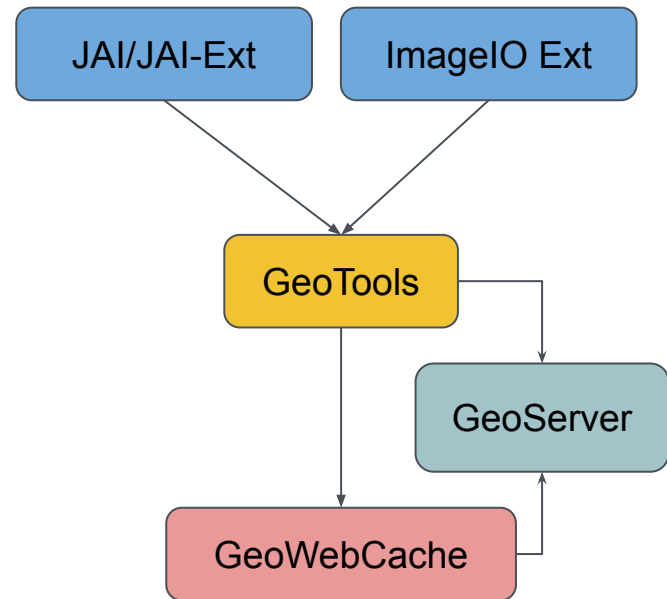
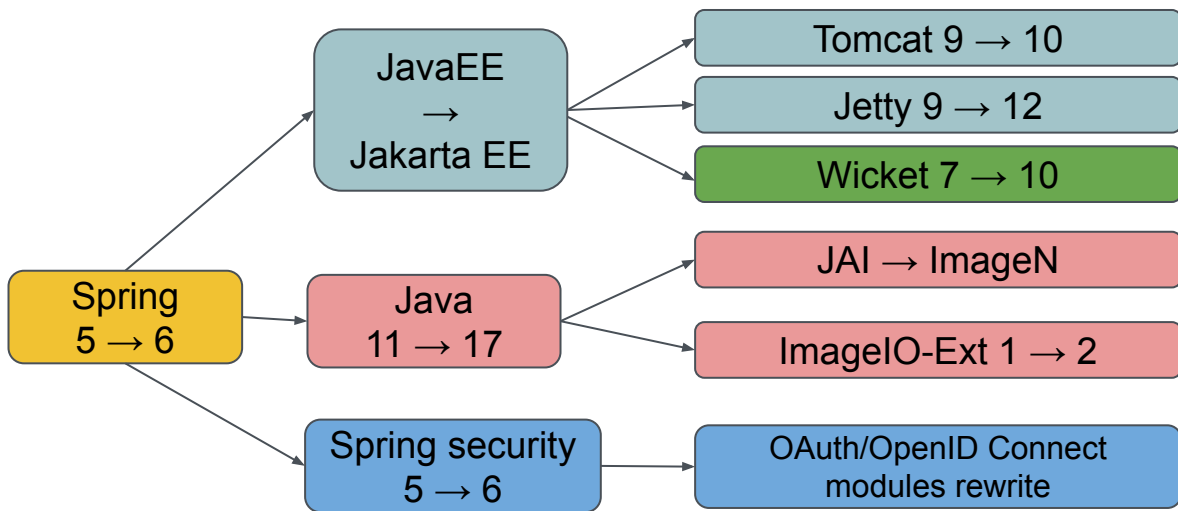


Java Advanced Imaging to ... ImageN

- Java Advanced Imaging is not open source... how to work with Java 17?
- Jody had Oracle donate its sources to the Eclipse Foundation under Apache license
- Rebranded as [ImageN](#)
- Fix that one and switch ImageIO-Ext / GeoTools / GeoServer to use it
- It's a lot of work! Massive technical risk! (probably the scariest roadmap item)
 - Donation included docs but not tests!
- Combined with GeoSolutions JAI-Ext
 - improved performance
 - provided test coverage!



The Java upgrade avalanche.... hits the entire stack!



Initial community response: Wicket Updates

Wicket 7→8

- **Migration:** Single developer (Brad Hards) started work in 2024!
 - Community lacked capacity to review and address feedback
 - Peter and Jody shared a test plan, noting what screens have been changed
 - However very difficult to provide instructions on how to find each screen
- GeoServer 2.25.x:
 - Wicket 8 missed the release window
 - Work was continued in Draft for 2.26.x

Wicket 8→9

- **Migration:** Wicket 9 update
- **CSP:** Content Security Policy headers
 - Restrictions to mitigate vulnerabilities
 - Combo of community testing
- **Dialog:** Wicket 9 deprecates “dialog”
 - Wicket 10 drops completely!
 - Asking us to roll our own!
 - Thanks to David for implementing this
- GeoServer 2.27.0:
 - Delayed release to get Wicket dialog testing: release candidate → nightly build
 - CSP testing was laborious, public release very valuable to stabilizing this work

Company consortium - Call for Crowdfunding!

Roadmap challenge did not update GeoServer
“in time” for Spring Framework 5.3 EOL!

Three companies stood up and started organizing the upgrade, contacting their customers and the larger community:

- More work than any one company!
- More funding than any one customer :)
- A detailed plan and target funding
- Collecting interested parties
- Regular meetings, progress updates

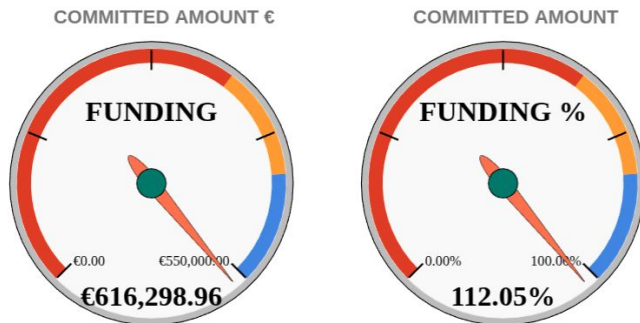


camptocamp

GEOCAT

GeoServer 3 Crowdfunding

- Funding secured! Took several months of relentless work from the 3 companies
- Some of the work as in-kind donations



GeoServer 3 is supported by the following organizations:



Individual donations: Abhijit Gujar, Hennessy Becerra, Ivana Ivanova, John Bryant, Jason Horning, Jose Macchi, Peter Smythe, Sajjadul Islam, Sebastiano Meier, Stefan Overkamp.

So many changes...

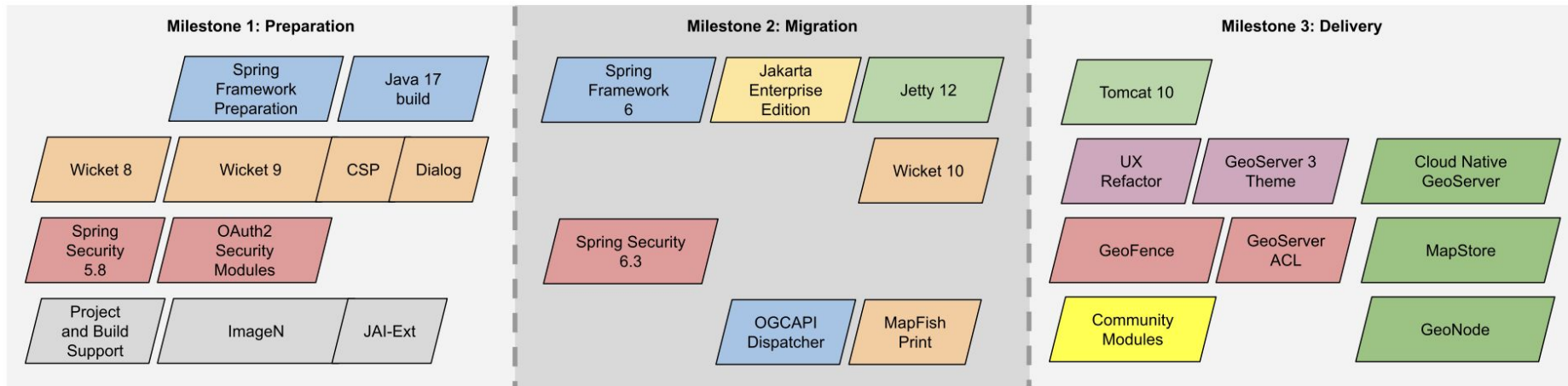
- We never attempted such a large change
- But it must be done to ensure GeoServer future
- So many changes deserve a new name



GeoServer³

Project Plan and Execution

GeoServer 3: Project Plan



May-Sept 2025

Oct-Dec 2025

Jan-Mar 2026



M1 preparation: nearing the precipice

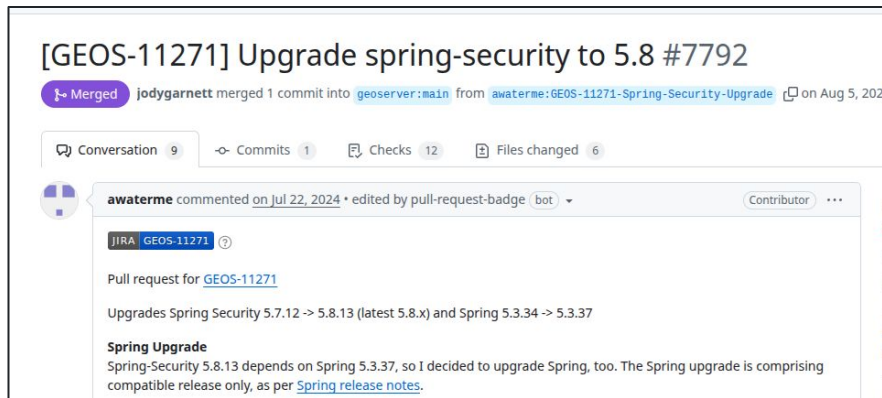
- For M1 we got as close as possible to the upgrades ... without breaking anything
- Any change we perform still works in Java 11 and Spring Framework 5
- Results collected in GeoServer series:
 - 2.27.0
 - 2.28.0



M1: Spring Security Prep!

- **Spring Security 5.8 upgrade**
 - Getting ready for Spring Security 6!
 - Upgrading to the latest Spring Security 5.8
 - Provides access to “new” OAuth2 client
- Provided in-kind by Andreas Watermayer

- **OAuth Community Modules**
 - Keycloak and OAuth cannot upgrade
 - With spring-security-5.8 we started rewriting from as new **OIDC** module



[GEOS-11271] Upgrade spring-security to 5.8 #7792

Merged jodygarnett merged 1 commit into geoserver:main from awaterme:GEOS-11271-Spring-Security-Upgrade on Aug 5, 2024

Conversation 9 Commits 1 Checks 12 Files changed 6

awaterme commented on Jul 22, 2024 • edited by pull-request-badge bot Contributor

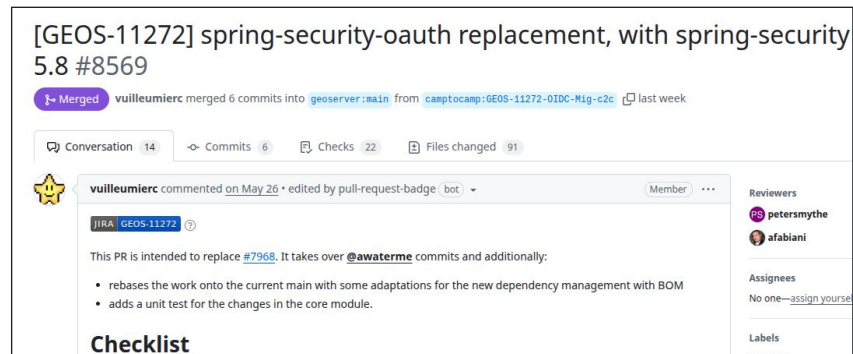
JIRA GEOS-11271

Pull request for GEOS-11271

Upgrades Spring Security 5.7.12 -> 5.8.13 (latest 5.8.x) and Spring 5.3.34 -> 5.3.37

Spring Upgrade

Spring-Security 5.8.13 depends on Spring 5.3.37, so I decided to upgrade Spring, too. The Spring upgrade is comprising compatible release only, as per [Spring release notes](#).



[GEOS-11272] spring-security-oauth replacement, with spring-security 5.8 #8569

Merged vuilleumierc merged 6 commits into geoserver:main from camptocamp:GEOS-11272-OIDC-Mig-c2c last week

Conversation 14 Commits 6 Checks 22 Files changed 91

vuilleumierc commented on May 26 • edited by pull-request-badge bot Member

JIRA GEOS-11272

This PR is intended to replace #7968. It takes over @awaterme commits and additionally:

- rebases the work onto the current main with some adaptations for the new dependency management with BOM
- adds a unit test for the changes in the core module.

Checklist

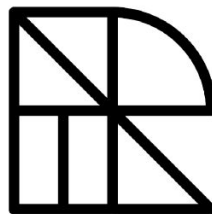
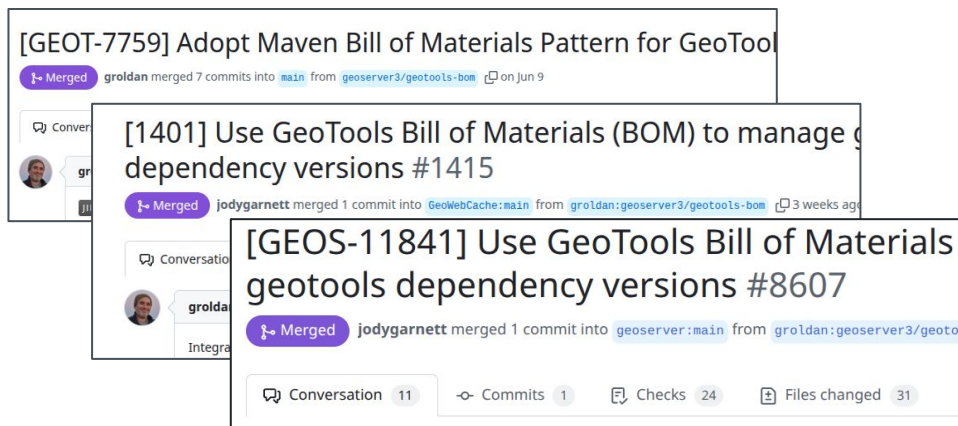
Reviewers: petersmythe, afabiani

Assignees: No one—assign yourself

Labels

M1: Java Baseline Prep

- **Centralized Bill of Materials**
- Manage dependent libraries in one file
- Single BOM file used by GeoTools, GeoWebCache, GeoServer
- **Java 17 as the minimum**
- Massive changes landed in July 2025!
- Used OpenRewrite automated migrations



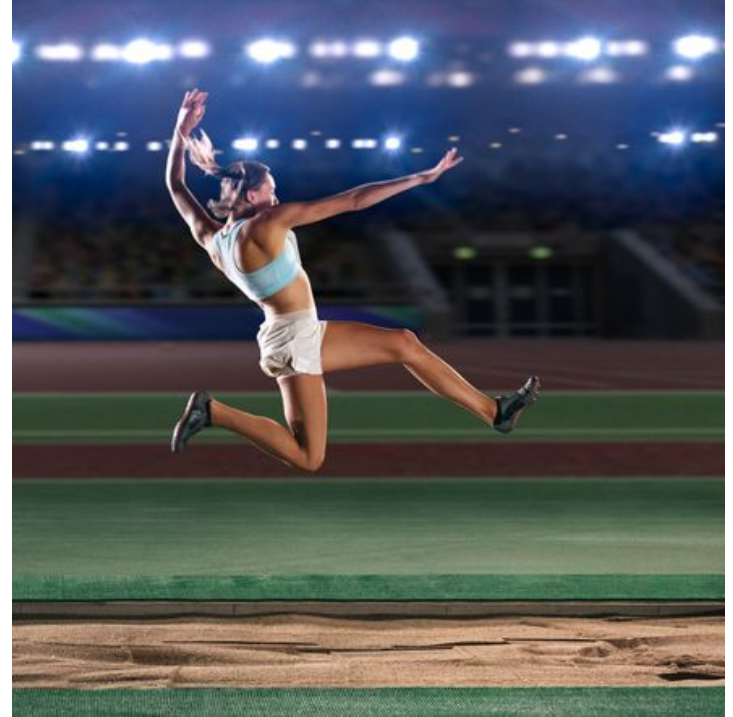
M1: ImageN

- Sources of JAI as donated to the Eclipse Foundation as a “new” **ImageN** project!
 - Modified to work with Java 17
 - Reduce code synchronization, increasing scalability
 - Removing “finalizers” (long standing issue)
 - Caution: No tests provided!
- JAI raster operators are pluggable, GeoServer/GeoTools actually use **JAI-EXT**
 - Bug fixes, Performance improvements
 - NoDATA support
 - Guess what? JAI-EXT has tests!
- Solution:
 - Merge JAI-EXT into ImageN
 - Setup ImageN with JAI-EXT operators (isolate old operators into a legacy module)



M2: here goes nothing!

- All of the changes to be done in one shot
 - All the disruptive changes in a single go
 - If we tried to do it one by one, nothing would compile anyways
- October 20th 2025, Code Sprint at the GeoSolutions offices



M2: Code Sprint

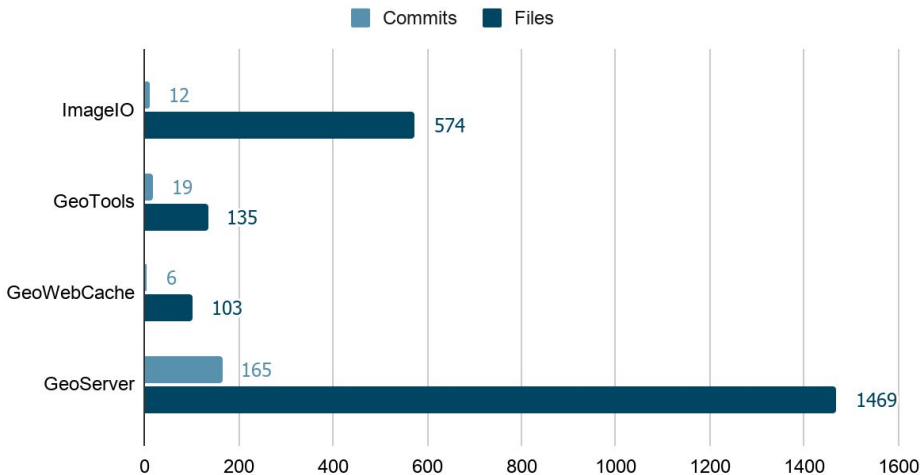
- “OpenRewrite” migrations run by Andrea Oct Fri 17th
- Monday Oct 20th → Saturday 26th
- Participants by company
 - CampToCamp: Gabriel, Cecile
 - GeoSolutions: Alessio, Andrea, Daniele
 - GeoCat: Jody, Ian
- One intense week, from “nothing works” to full build passing (wildly successful outcome!)



M2: The code sprint in numbers

- ImageIO-Ext (XML parsing migration)
 - 12 commits, 574 files changed, 57314 insertions(+), 65460 deletions(-)
- GeoTools (XML parsing, HTTP client upgrade)
 - 19 commits, 135 files changed, 1910 insertions(+), 1059 deletions(-)
- GeoWebCache (XML parsing, Jakarta EE, Spring upgrade):
 - 6 commits, 103 files changed, 562 insertions(+), 513 deletions(-)
- GeoServer
 - **165 commits, 1469 files changed, 8045 insertions(+), 27208 deletions(-)**

Milestone 2 Code Sprint



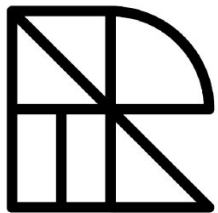
M2: Milestone 2 Sprint Success

- Jumped and landed!
- GeoServer core and extensions module build
- Community modules compile
- GeoServer 3 starts up and “mostly works” in embedded jetty
- General development can resume “Code Freeze” over!



M3: *Surprise!* Spring 7 upgrade

- Spring release version 7 November 13
- Along with the announcement we noticed they drop Spring 6 support in a few months!
- ... so we upgraded to Spring 7



spring®

M3: User Interface Theme: Maintainability

Both Tailwind and Bootstrap too error prone:

- CSS expert “cleaned up” **geoserver.css** with clear examples for devs to cut and paste
- New gs-theme module allows geoserver.css use for Wicket, OpenLayers preview, and Freemarker templates used by OGCAPI

Layer Preview

List of all layers configured in GeoServer and provides previews in various formats for each.

Results 1 to 25 (out of 27 items)				
Search				
Type	Title	Name	Common Formats	All Formats
	Boundary lines (Disputed)	ne:boundary_lines_disputed	OpenLayers GML KML	Select one
	Boundary lines	ne:boundary_lines_land	OpenLayers GML KML	Select one
	Boundary lines (Territories)	ne:boundary_lines_map_units	OpenLayers GML KML	Select one
	Boundary lines (Maritime)	ne:boundary_lines_maritime	OpenLayers GML KML	Select one

```
:root,
[data-gs-theme=light] {
  --gs-primary: #0076a1;
  --gs-brand-gradient: linear-gradient(to r
  --gs-secondary: #6c757d;
  --gs-success: #198754;
  --gs-info: #2cb5e8;
  --gs-warning: #ffc107;
  --gs-danger: #dc3545;
  --gs-light: #f8f9fa;
  --gs-dark: #212529;
  --gs-primary-rgb: 0, 118, 161;
  --gs-secondary-rgb: 108, 117, 125;
  --gs-success-rgb: 25, 135, 84;
  --gs-info-rgb: 44, 181, 232;
  --gs-warning-rgb: 255, 193, 7;
  --gs-danger-rgb: 220, 53, 69;
  --gs-light-rgb: 248, 249, 250;
  --gs-dark-rgb: 33, 37, 41;
  --gs-primary-text-emphasis: #004761;
  --gs-secondary-text-emphasis: #2b2f32;
  --gs-success-text-emphasis: #0a3622;
  --gs-info-text-emphasis: #004d66;
  --gs-warning-text-emphasis: #664d03;
  --gs-danger-text-emphasis: #58151c;
  --gs-light-text-emphasis: #495057;
  --gs-dark-text-emphasis: #495057;
```

GeoServer CSS - Component Showcase

Visual reference for main components defined in [geoserver.css](#).

Theme: Light Dark

1. Color Palette

CSS custom properties defined on `:root` (and `[data-gs-theme=dark]` override).

Primary Secondary Success Info Warning Danger Light Dark

```
/* In CSS */
color: var(--gs-primary);
background: var(--gs-danger-bg-subtle);
border-color: var(--gs-border-color);
```

2. Typography

Heading 1
Heading 2
Heading 3
Heading 4
Heading 5

Body paragraph - [link example](#), **bold**, *italic*, `inline code`, **highlighted**.

Small text (.875em) Extra-small (.675em)

```
<h1>Heading 1</h1>
<p>Body text with <a href="#">link</a>, <strong>bold</strong>, <code>code</code>=</p>
<span class="text-xs">Extra-small</span>
```

3. Buttons

Default Primary Danger Outline Primary Outline Danger Disabled

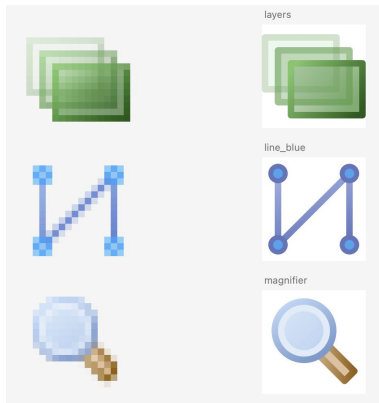
```
<button class="button">Default</button>
<button class="button primary">Primary</button>
<button class="button danger">Danger</button>
<button class="button outline-primary">Outline Primary</button>
<button class="button outline-danger">Outline Danger</button>
<button class="button primary" disabled>Disabled</button>
```

<!-- .button class works on <a too -->
Link Button

M3: User Interface Theme: Icons

Hunted down the original “GeoSilk” icons:

- Lots of examples of SVG or upscaling the initial 16 x 16 “Silk” icon set.
- Reuse where possible, rework when we must
- Theme can smooth change to SVG as additional icons ported



14. Icons (.gs-icon)

All `.gs-icon` classes parsed live from `geoserver.css`. Use as `<i class="gs-icon gs-icon-{name}"></i>`.

Standalone



Inside button



Primary + icon

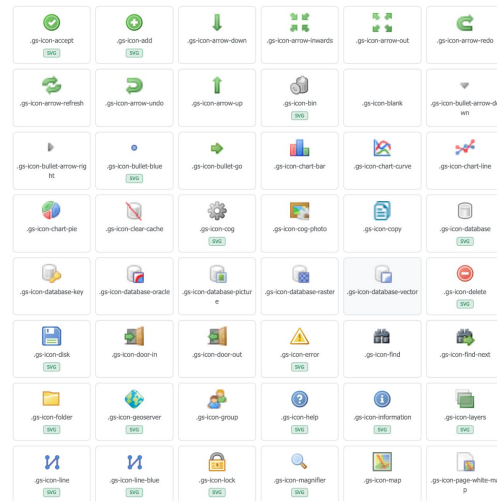


Danger + icon

```
<!-- Standalone icon -->
<i class="gs-icon gs-icon-accept"></i>

<!-- Icon inside a button -->
<button ><i class="gs-icon gs-icon-accept"></i> Accept</button>
<button class="primary"><i class="gs-icon gs-icon-add"></i> Add</button>
<button class="danger"><i class="gs-icon gs-icon-delete"></i> Delete</button>
```

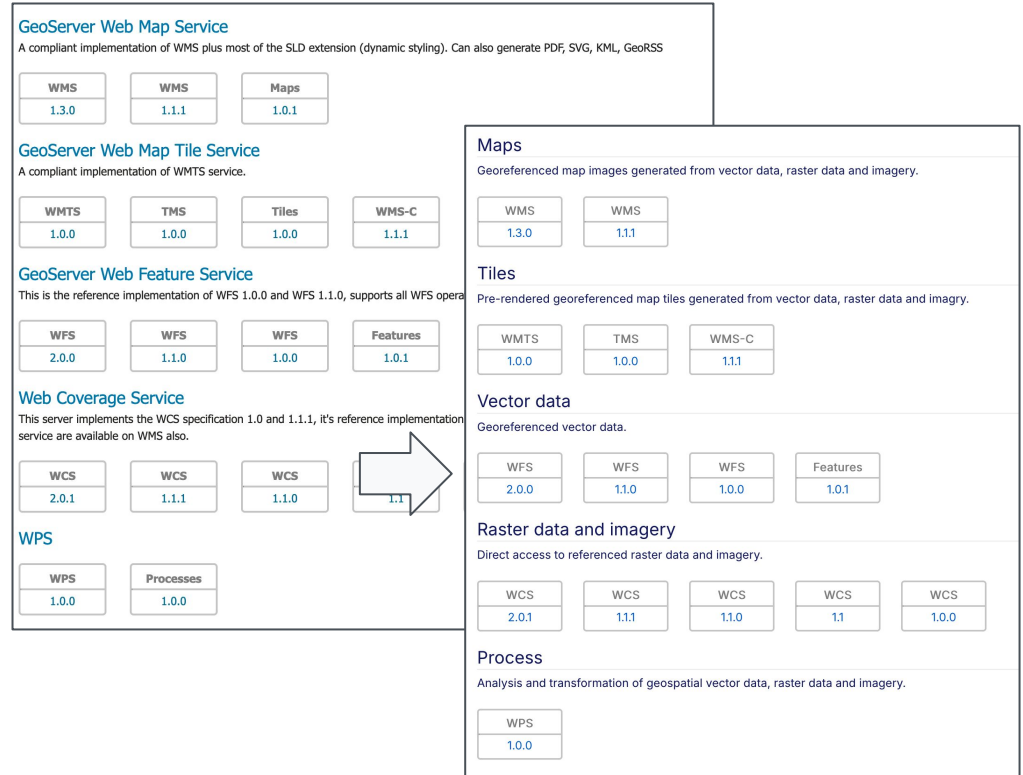
Filter icons...



M3: User Experience

Looking ahead to introduction of OGCAPI Services:

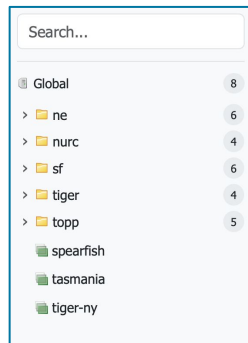
- Group services by Category
 - Renaming the Services Pages
 - Responsible for multiple services



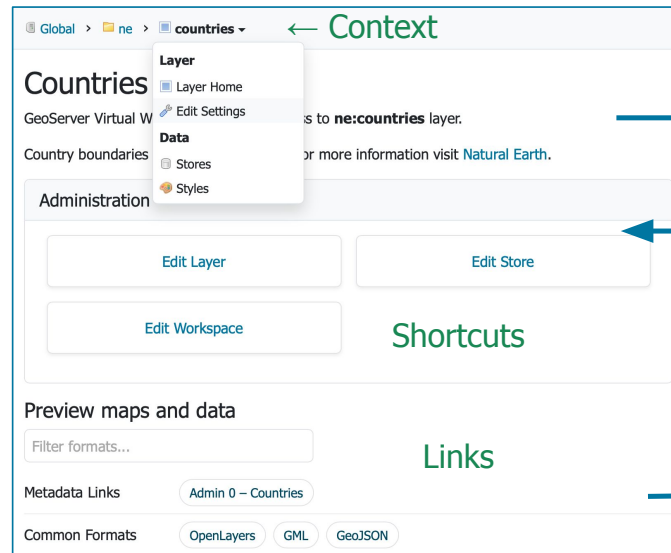
M3: *Surprise!* Context-driven workflow

Step 1

Search →
Results →



Step 2

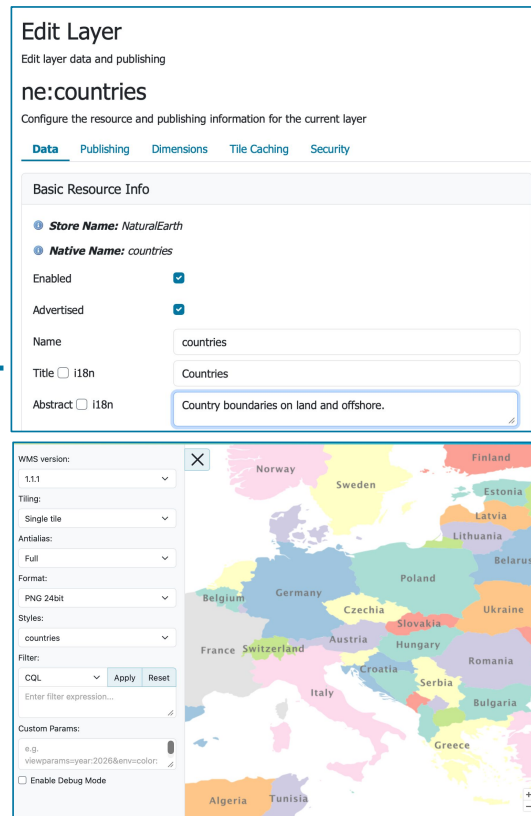


Edit

Save

Preview

Step 3



M3: Community modules

- Push on with testing of community modules
- Migrate as possible to the new pending profile
- Dropped: *taskmanager* , *notification* , *spatialjson* , *wps-remote*

- [Backup and Restore Documentation](#)
- [COG \(Cloud Optimized GeoTIFF\) Documentation](#)
- [Dynamic colormap generation](#)
- [CoverageJSON output format](#)
- [DDS/BIL\(World Wind Data Formats\) Extension](#)
- [Elasticsearch data store](#)
- [Features-Autopopulate Extension](#)
- [Features-Templating Extension](#)
- [WFS FlatGeobuf input and output format](#)
- [GDAL based WCS Output Format](#)
- [GeoPackage Extension](#)
- [GeoServer Access Control List authorization](#)
- [GeoStyler](#)

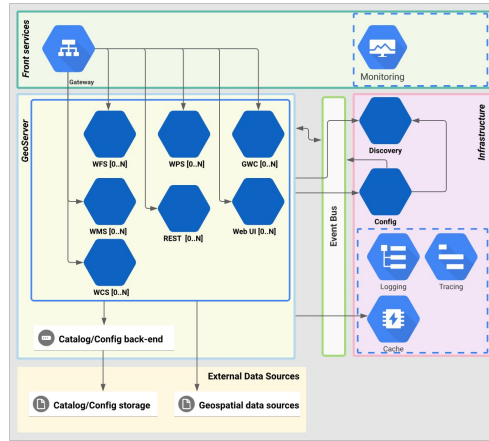
- [Graticule Extension](#)
- [GSR Extension](#)
- [GWC Azure BlobStore plugin](#)
- [GWC Google Cloud Storage BlobStore plugin](#)
- [GWC MBTiles layer plugin](#)
- [GWC SQLite Plugin](#)
- [SAP HANA](#)
- [Hazelcast Clustering Plugin](#)
- [Overview](#)
- [Simple clustering cache invalidation setup](#)
- [Enabling Wicket session sharing](#)
- [Importer JDBC storage](#)
- [JDBCConfig](#)
- [JDBCStore](#)

- [JMS based Clustering](#)
- [JWT Headers](#)
- [Libdeflate](#)
- [MBTiles Extension](#)
- [Monitoring Kafka storage](#)
- [Monitoring with Micrometer support](#)
- [ncWMS WMS extensions support](#)
- [GHRSSST NetCDF output](#)
- ~~[Notification community module Plugin Documentation](#)~~
- [OGC API modules](#)
- [OGR datastore](#)
- [OAuth2 OpenID Connect](#)
- [PMTiles DataStore](#)
- [Proxy Base Extension](#)
- ~~[WPS Remote community module](#)~~
- [S3 Support for GeoTiff](#)
- [Schemaless Features Mongo Plugin](#)
- [SingleStore](#)
- [Smart Data Loader Extension](#)
- ~~[SpatialJSON WFS Output Format Extension](#)~~
- [STAC Datastore extension](#)
- [SQLR data store](#)
- ~~[Task Manager](#)~~
- [Vector Mosaic datastore](#)
- [VSI Virtual File System Support](#)
- [HTTP Based Authorization plug-in](#)
- [WMS WebP output format](#)
- [WFS FreeMarker Extension](#)
- [WPS Download NetCDF](#)
- [WPS longitudinal profile process](#)
- [WPS OpenAI process](#)

M3: Downstream Integrations

- **GeoServer Cloud**
- Cloud-native, microservices, distribution of GeoServer
- Spring Cloud 2025.1 / Spring Boot 4 / Spring Framework 7 on Java 25

- **GeoServer ACL**
- Access Control List
- Security module for GeoServer Cloud
- Donated back to the GeoServer project



M2: Testing and Quality Assurance

- **Dependabot automatic update PRs**
- Evaluate and merge PRs that are green
- Backwards incompatible upgrades are still a problem

✓ All checks have passed
49 successful checks

✓	QA GitHub CI / QA (pull_request)	Successful in 24m
✓	Run Acceptance Tests / acceptance (17) (pull_request)	Success
✓	Run Acceptance Tests / acceptance (21) (pull_request)	Success
✓	Run Acceptance Tests / acceptance (25) (pull_request)	Success
✓	Run Acceptance Tests / Build GeoServer WAR (pull_request)	Success
✓	Run CITE Tests / Build GeoServer WAR (pull_request)	Success
✓	Run CITE Tests / CITE (geotiff11, 17) (pull_request)	Successful
✓	Run CITE Tests / CITE (geotiff11, 21) (pull_request)	Successful



- **OGC compliance tests (CITE)**
Run for WMS, WFS, WCS, WMTS, OGC API Features/Tiles, GeoTIFF, GeoPackage
- **Assembly tests**
Startup each extension/community module
- **Acceptance tests**
Common REST API / OGC service workflow

Done for each Pull Request we receive !

GSIP-221: Documentation migrated to Markdown

Migration of docs:

- RST → Markdown
- Using pandoc and regex!

New screenshots:

- AfriGIS lead review
And collection of new
ScreenShots

The screenshot shows the GeoServer Documentation website. The top navigation bar includes the GeoServer logo, 'Documentation', a dropdown for 'main (Latest)', and links for 'HOME', 'USER MANUAL', 'DEVELOPER GUIDE', 'DOCUMENTATION GUIDE', and 'API REFERENCE'. A search bar is on the right. The left sidebar shows the 'USER MANUAL' with a list of topics: Introduction, Installation, Getting started, Using the web administration interface, Publishing a GeoPackage (highlighted), Publishing a GeoTIFF, Publishing a Layer Group, Publishing a style, Preflight Checklist, and Publishing a shapefile. The main content area is titled 'Publishing a GeoPackage' and contains a 'Note' box stating: 'This tutorial assumes that GeoServer is running at `http://localhost:8080/geoserver`'. Below the note, the section 'Data preparation' begins with the text 'First let's gather the data that we'll be publishing.' followed by a numbered list: 1. The sample data folder includes `data/ne/natural_earth.gpkg`; 2. This file contains small scale 1:110m data:

- [coastlines](#)
- [countries](#)
- [boundary lines](#)

On the right side of the main content area, there is a 'PAGE CONTENTS' section with a list of links: Data preparation, Creating a new workspace, Create a store, Creating a layer, and Previewing the layer.

M3: GeoServer 3.0 delivered!

- Starts up and works in Tomcat, Jetty, Docker, Installers
- All core and extensions modules work
- Passes OGC compliance tests
- New OIDC modules work properly
- Transparently read and upgrade from a GeoServer 2.x data directory
- Refreshed UI Theme
- Context-Driven User Experience
- No known problematic dependencies (the day of the release, situation is very fluid)

GeoServer 3.0.0 Release

Jun 11, 2026 · Jody Garnett

GeoServer 3.0.0 release is now available with downloads [extensions](#).

This is a stable release of GeoServer 3.0.x series, GeoServer 3.0, and GeoWebCache 2.0.0.

Thanks to Andrea Aime (GeoSolutions), Jody Garnett (GeoSolutions), and the community for their contributions to this release.

Security Considerations

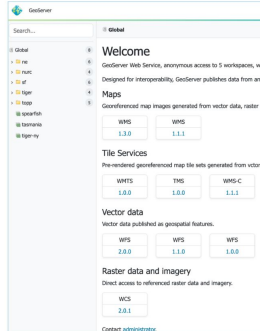
This release addresses security vulnerabilities and is an important update for all users.

- GEOS-12043 CVE-2025-27511 JNDI Vulnerability
- GEOS-11920 CVE-2025-58175 Server-Side Request Forgery
- GEOS-11918 CVE-2025-52465 Arbitrary file write
- GEOS-11777 CVE-2024-45747 Server-Side Template Injection

The use of the CVE system allows the GeoServer team to track and address security vulnerabilities. See project [security policy](#) for more information on how we handle security.

Welcome to GeoServer 3

We are overjoyed to share the initial release of GeoServer 3.0.0, a long road, a year of development, and a lot of planning with the organizations and individuals supporting GeoServer 3.0.



GeoServer 3.0.0

Released on June 11, 2026

[DOI: 10.5281/zenodo.155281/zenodo](#)

[Changelog](#)
GeoServer 3.0.0 release notes

[Announcement](#)
GeoServer blog.

Packages

- Platform Independent Binary
Operating system independent runnable binary.
- Web Archive
Web Archive (war) for servlet containers.
- Windows Installer
Installer for Windows platforms.
- Data Directory
GeoServer Data directory with sample layers.

Documentation

- User Guide HTML
Export of GeoServer user guide.
- Javadoc
API documentation.

Source Code

- zip | tar.gz
Source code packages for tag 3.0.0.
- GitHub
Browse source code on GitHub.

Extensions

Extensions
GeoServer Extension downloads.

Vector Formats

- App Schema
- DB2
- MongoDB
- MySQL
- Oracle
- Pregeneralized Features
- SQL Server

Cartography

- Chart Symbolizer
- CSS Styling
- MBStyle Styling
- Printing
- YSLD Styling

Miscellaneous

- Control Flow
- Cross Layer Filtering
- GWC S3 tile storage
- IAU
- Importer
- INSPIRE
- Metadata
- Monitor (Core)
- Request parameters extractor
- Resource Browser Tool

Security

- Key authentication
- CAS
- OIDC

Coverage Formats

- ArcGrid
- GDAL
- GRIB
- Image Pyramid
- JPEG2K
- NetCDF
- Raster Attribute Table
- WorldImage

Output Formats

- DXF
- Excel
- GeoPkg
- Image Map
- JPEG Turbo
- MapML
- NetCDF
- OGR (WFS, WPS)
- Vector Tiles
- XSLT

OGC APIs

- Features

OGC Services

- CSW
- CSW ISO Metadata Profile
- WCS 1.0
- WCS 1.1
- WCS 2.0 EO
- WPS
- WPS Clustering (Hazelcast, JDBC)
- WPS download
- WMTS multi-dimensional

Web Services

- SLDService

M4: To be continued...

- **Downstream integration testing**
- GeoFence: Update to Spring Framework 7
- GeoNode
- MapStore



- **General cleanup and Docs**
 - Formattin, glitches, bullet point level
 - Screenshots to update
 - Install instructions, version numbers

Please help, use the edit button!

- **User Interface**
 - Help add SVG icons!
 - Sensible headings for service pages (Cartography, Download Formats,...)

Please help, improves usability!

Documentation

Documentation migrated to Markdown

Same content

New UI experience

Fast Search

The screenshot displays the GeoServer Documentation website. The top navigation bar includes the GeoServer logo, the word 'Documentation', a dropdown menu for 'main (Latest)', and a search bar. Below this is a secondary navigation bar with links for 'HOME', 'USER MANUAL', 'DEVELOPER GUIDE', 'DOCUMENTATION GUIDE', and 'API REFERENCE'. The main content area is titled 'Publishing a GeoPackage' and includes a 'Note' box stating that the tutorial assumes GeoServer is running at `http://localhost:8080/geoserver`. The left sidebar shows the 'USER MANUAL' menu with links to 'Introduction', 'Installation', 'Getting started', 'Using the web administration interface', 'Publishing a GeoPackage' (highlighted), 'Publishing a GeoTIFF', 'Publishing a Layer Group', 'Publishing a style', 'Preflight Checklist', and 'Publishing a shapefile'. The right sidebar shows the 'PAGE CONTENTS' for the current page, listing 'Data preparation', 'Creating a new workspace', 'Create a store', 'Creating a layer', and 'Previewing the layer'.

GeoServer Documentation main (Latest) geoserver/geoserver 3.0.0 ☆ 4.4k 2.3k

HOME USER MANUAL DEVELOPER GUIDE DOCUMENTATION GUIDE API REFERENCE

USER MANUAL

- Introduction
- Installation
- Getting started
 - Using the web administration interface
 - Publishing a GeoPackage**
 - Publishing a GeoTIFF
 - Publishing a Layer Group
 - Publishing a style
 - Preflight Checklist
 - Publishing a shapefile

Publishing a GeoPackage

This tutorial walks through the steps of publishing a GeoPackage with GeoServer.

Note

This tutorial assumes that GeoServer is running at `http://localhost:8080/geoserver`.

Data preparation

First let's gather the data that we'll be publishing.

1. The sample data folder includes `data/ne/natural_earth.gpkg`
2. This file contains small scale 1:110m data:
 - [coastlines](#)
 - [countries](#)
 - [boundary lines](#)

PAGE CONTENTS

- Data preparation
- Creating a new workspace
- Create a store
- Creating a layer
- Previewing the layer

The documentation dark side

Look for the “sun” button
and eclipse all that white



The screenshot shows the GeoServer documentation website in dark theme. The top navigation bar includes 'GeoServer', 'Documentation', and a dropdown for 'main (Latest)'. The main content area is titled 'Publishing a GeoPackage' and includes a 'Note' box stating: 'This tutorial assumes that GeoServer is running at <http://localhost:8080/geoserver>.' The left sidebar lists the 'USER MANUAL' sections, and the right sidebar shows the 'PAGE CONTENTS' for the current page.

USER MANUAL

- Introduction
- Installation
- Getting started
 - Using the web administration interface
 - Publishing a GeoPackage
 - Publishing a GeoTIFF
 - Publishing a Layer Group
 - Publishing a style
 - Preflight Checklist
 - Publishing a shapefile

Publishing a GeoPackage

This tutorial walks through the steps of publishing a GeoPackage with GeoServer.

Note

This tutorial assumes that GeoServer is running at <http://localhost:8080/geoserver>.

Data preparation

First let's gather the data that we'll be publishing.

1. The sample data folder includes `data/ne/natura1_earth.gpkg`
2. This file contains small scale 1:110m data:
 - [coastlines](#)
 - [countries](#)
 - [boundary lines](#)

PAGE CONTENTS

- Data preparation
- Creating a new workspace
- Create a store
- Creating a layer
- Previewing the layer

New GeoServer 3 Theme

Mobile/small screen friendly

GeoServer Data Services Server Tile Cache Security Utilities Monitor

Search in topp...

+ New

Global 8

topp 5

- states
- tasmania_cities
- tasmania_roads
- tasmania_state_boundaries
- tasmania_water_bodies

Layers

Manage the layers being published by GeoServer.

[Add a new layer](#) [Remove selected layers](#)

<< < 1 > >> Results 1 to 5 (out of 5 items)

Search

☐	Type	Title	Name	Store	Enabled
☐	USA Population	topp:states	states_shapefile	✓	
☐	Tasmania cities	topp:tasmania_cities	taz_shapes	✓	
☐	Tasmania roads	topp:tasmania_roads	taz_shapes	✓	
☐	Tasmania state boundaries	topp:tasmania_state_boundaries	taz_shapes	✓	
☐	Tasmania water bodies	topp:tasmania_water_bodies	taz_shapes	✓	

<< < 1 > >> Results 1 to 5 (out of 5 items)



en

Search in topp...

Global > topp

Layers

Manage the layers being published by GeoServer.

[Add a new layer](#) [Remove selected layers](#)

<< < 1 > >> Results 1 to 5 (out of 5 items)

Search

☐

Type

Title USA Population

Name topp:states

Store states_shapefile

Enabled ✓

Native SRS EPSG:4326

☐

Type

Title Tasmania cities

Name topp:tasmania_cities

Store taz_shapes

Enabled ✓

Native SRS EPSG:4326

☐

Type

Title Tasmania roads

Name topp:tasmania_roads

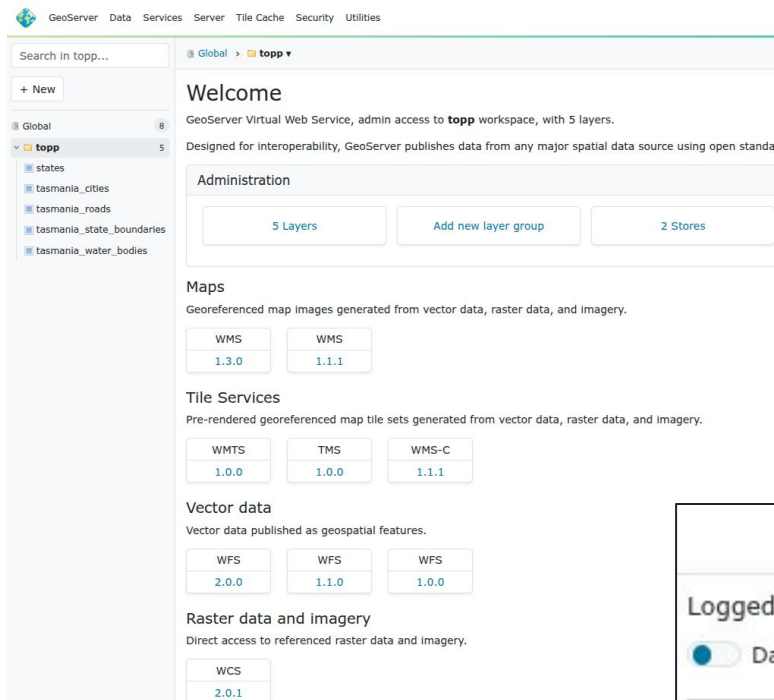
Adaptive two-column form layout

The screenshot shows the GeoServer web interface. The top navigation bar includes links for GeoServer, Data, Services, Server, Tile Cache, Security, and Utilities. A search bar and a '+ New' button are on the left. A sidebar on the left lists various data stores: Global (8), ne (6), nurc (4), sf (6), tiger (4), topp (5), spearfish, tasmania, and tiger-ny. The main content area is titled 'Global' and has tabs for 'Service' and 'WMS'. The 'WMS' tab is active, showing 'KML Options'. The form is organized into two columns. The left column contains 'Default Reflector Mode' (dropdown set to 'refresh'), 'Default Superoverlay Mode' (dropdown set to 'auto'), 'Generate vector placemarks (KMATTR)' (checkbox checked), 'Generate raster placemarks (kmlplacemark)' (checkbox unchecked), and 'Raster/vector threshold (0-100, default 40)' (text input set to '40'). The right column is empty.



This is a zoomed-in view of the 'WMS' configuration page. It shows the 'KML Options' section with the following settings: 'Default Reflector Mode' is 'refresh', 'Default Superoverlay Mode' is 'auto', 'Generate vector placemarks (KMATTR)' is checked, 'Generate raster placemarks (kmlplacemark)' is unchecked, and 'Raster/vector threshold (0-100, default 40)' is '40'. Below this is the 'Resource consumption limits' section with the following settings: 'Max rendering memory (KB)' is '65536', 'Max rendering time (s)' is '60', 'Max rendering errors (count)' is '1000', and 'Max number of dimension values' is '100'.

Join the dark side of the administration! (optional)



GeoServer Data Services Server Tile Cache Security Utilities

Search in topp...

+ New

Global 8

topp 5

- states
- tasmania_cities
- tasmania_roads
- tasmania_state_boundaries
- tasmania_water_bodies

Welcome

GeoServer Virtual Web Service, admin access to **topp** workspace, with 5 layers.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards.

Administration

5 Layers Add new layer group 2 Stores

Maps

Georeferenced map images generated from vector data, raster data, and imagery.

WMS 1.3.0 WMS 1.1.1

Tile Services

Pre-rendered georeferenced map tile sets generated from vector data, raster data, and imagery.

WMTS 1.0.0 TMS 1.0.0 WMS-C 1.1.1

Vector data

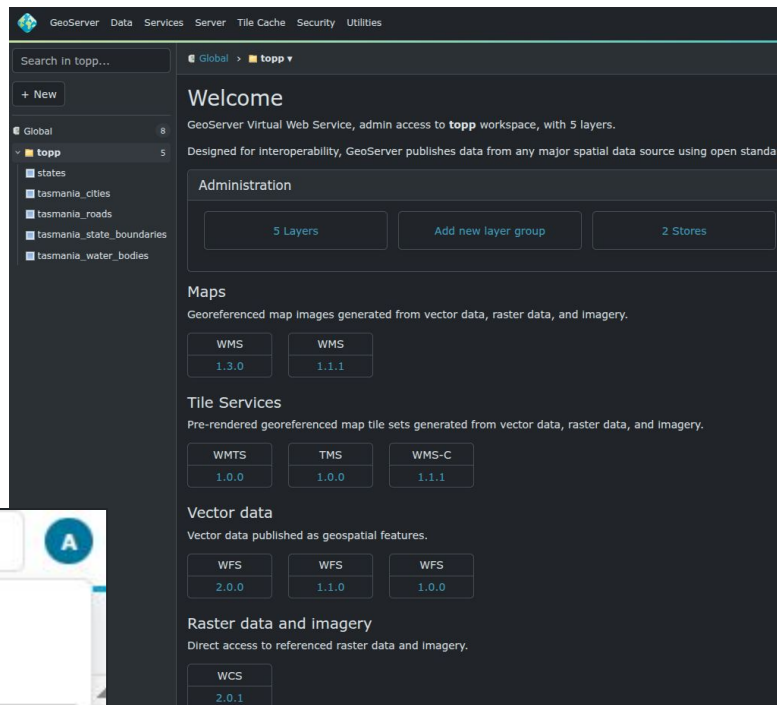
Vector data published as geospatial features.

WFS 2.0.0 WFS 1.1.0 WFS 1.0.0

Raster data and imagery

Direct access to referenced raster data and imagery.

WCS 2.0.1



GeoServer Data Services Server Tile Cache Security Utilities

Search in topp...

+ New

Global 8

topp 5

- states
- tasmania_cities
- tasmania_roads
- tasmania_state_boundaries
- tasmania_water_bodies

Welcome

GeoServer Virtual Web Service, admin access to **topp** workspace, with 5 layers.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards.

Administration

5 Layers Add new layer group 2 Stores

Maps

Georeferenced map images generated from vector data, raster data, and imagery.

WMS 1.3.0 WMS 1.1.1

Tile Services

Pre-rendered georeferenced map tile sets generated from vector data, raster data, and imagery.

WMTS 1.0.0 TMS 1.0.0 WMS-C 1.1.1

Vector data

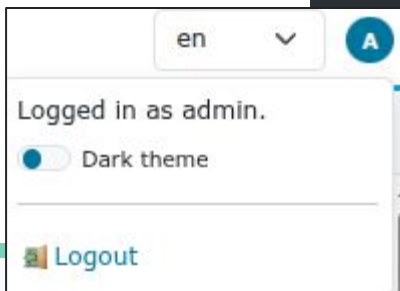
Vector data published as geospatial features.

WFS 2.0.0 WFS 1.1.0 WFS 1.0.0

Raster data and imagery

Direct access to referenced raster data and imagery.

WCS 2.0.1



en

Logged in as admin.

☒ Dark theme

Logout

New User Experience

Navigation Menu →

Search →

Results →

 GeoServer

Language → en 

Search...

Global 8

> ne 6

> nurc 4

> sf 6

> tiger 4

> topp 5

 spearfish

 tasmania

 tiger-ny

Global ←Context (ie breadcrumbs)

Welcome (Page)

GeoServer Web Service, anonymous access to 5 workspaces, with 28 layers.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards. For more information visit [OSGeo](#).

Maps

Georeferenced map images generated from vector data, raster data, and imagery.

WMS
1.3.0

WMS
1.1.1

Tile Services

Pre-rendered georeferenced map tile sets generated from vector data, raster data, and imagery.

WMTS
1.0.0

TMS
1.0.0

WMS-C
1.1.1

Vector data

Vector data published as geospatial features.

WFS
2.0.0

WFS
1.1.0

WFS
1.0.0

Raster data and imagery

Direct access to referenced raster data and imagery.

WCS
2.0.1

Contact [administrator](#).

←Login

Search

Use search to quickly locate:

- Workspace
- Layers
- Layer Groups

Results are shown as Tree:

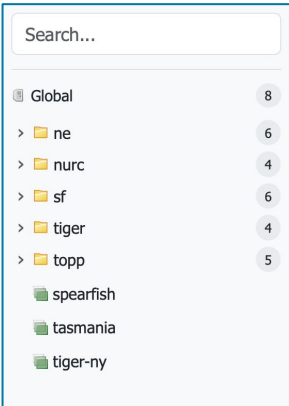
- Use the tree to explore results
- Click on entry to establish context
- Smoothly handles large catalogues

Typing autocomplete:

- Results update as you type
- Global and workspaces include count

Search →

Results →



← Top-Level

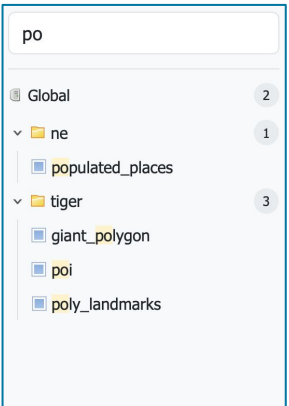
← Workspaces

← Layer Groups

The screenshot shows a search interface with a search bar at the top. Below it is a tree view of search results. The tree starts with a 'Global' entry with a count of 8. Under 'Global' are several workspace entries: 'ne' (6), 'nurc' (4), 'sf' (6), 'tiger' (4), and 'topp' (5). Under 'topp' are three layer group entries: 'spearfish', 'tasmania', and 'tiger-ny'. Green arrows point from the text labels to the corresponding parts of the tree: 'Top-Level' points to the 'Global' entry, 'Workspaces' points to the workspace entries, and 'Layer Groups' points to the layer group entries under 'topp'.

Search →

Results →

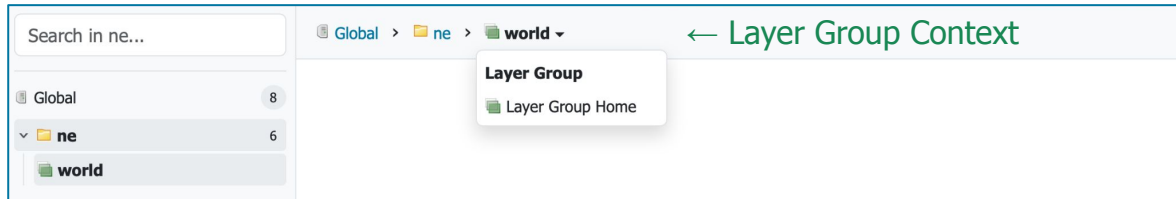
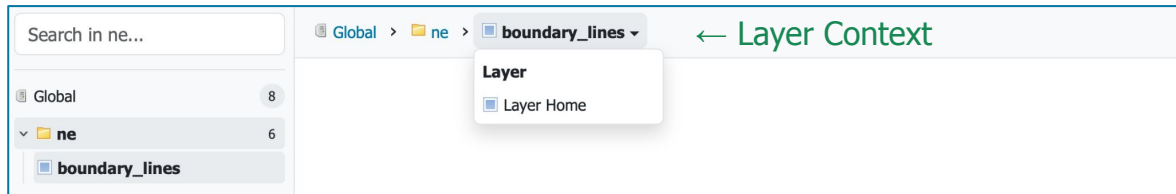
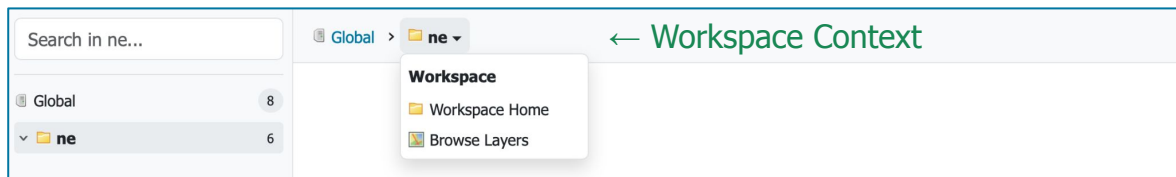


← Layers

The screenshot shows the same search interface but with the search bar containing the text 'po'. The tree view is filtered to show only results containing 'po'. The 'Global' entry now has a count of 2. Under 'Global' are two workspace entries: 'ne' (1) and 'tiger' (3). Under 'ne' is one layer entry: 'populated_places'. Under 'tiger' are three layer entries: 'giant_polygon', 'poi', and 'poly_landmarks'. A green arrow points from the text label 'Layers' to the layer entries in the tree.

Context

- Establish context by clicking on a search result
 - Global
 - Workspace
 - Layer
 - Layer Group
- Shown as “breadcrumb” along the top of the page
- Use drop-down context menu for quick actions



Global

Global context:

- Customize “Welcome” title and message, along with Contact Information
- Shows enabled Services along with their title and description

Global Services Disabled

 Global

Welcome

GeoServer, anonymous access. Select a workspace or layer, above, to change to a virtual web service.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards. For more information visit [OSGeo](#).

Contact [administrator](#).

Global Services Enabled

 Global

Welcome

GeoServer Web Service, anonymous access to 5 workspaces, with 28 layers.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards. For more information visit [OSGeo](#).

Maps

Georeferenced map images generated from vector data, raster data, and imagery.

WMS
1.3.0

WMS
1.1.1

Tile Services

Pre-rendered georeferenced map tile sets generated from vector data, raster data, and imagery.

WMTS
1.0.0

TMS
1.0.0

WMS-C
1.1.1

Vector data

Vector data published as geospatial features.

WFS
2.0.0

WFS
1.1.0

WFS
1.0.0

Raster data and imagery

Direct access to referenced raster data and imagery.

WCS
2.0.1

Contact [administrator](#).

← Title

← Welcome

← Services

← Contact

Workspace

Workspace context:

- Shows custom “Welcome” title and message along with contact information
- Shows enabled workspace virtual services along with their title and description

The screenshot shows the GeoServer workspace context for 'Natural Earth'. The breadcrumb navigation at the top reads 'Global > ne'. The main title is 'Natural Earth', followed by a description: 'GeoServer Virtual Web Service, anonymous access to ne workspace, with 6 layers.' Below this is a paragraph about Natural Earth being a public domain map dataset. The section 'GeoServer Natural Earth Maps' describes map images generated from Natural Earth vector data, raster data and imagery, and lists two WMS services with versions 1.3.0 and 1.1.1. The section 'GeoServer Natural Earth Tiles' describes map tiles generated from Natural Earth vector data, raster data and imagery, and lists three services: WMTS (1.0.0), TMS (1.0.0), and WMS-C (1.1.1). The section 'GeoServer Natural Earth Vector data' describes Natural Earth feature data, and lists three WFS services with versions 2.0.0, 1.1.0, and 1.0.0. The section 'GeoServer Natural Earth Raster data and imagery' describes direct access to Natural Earth raster data and imagery, and lists one WCS service with version 2.0.1.

Global > ne

Natural Earth

GeoServer Virtual Web Service, anonymous access to **ne** workspace, with 6 layers.

Natural Earth is a public domain map dataset available at 1:10m, 1:50m, and 1:110 million scales. Featuring tightly integrated vector and raster data, with Natural Earth you can make a variety of visually pleasing, well-crafted maps with cartography or GIS software. For more information visit [Natural Earth](#).

GeoServer Natural Earth Maps

Map images generated from Natural Earth vector data, raster data and imagery.

WMS	WMS
1.3.0	1.1.1

GeoServer Natural Earth Tiles

Map tiles generated from Natural Earth vector data, raster data and imagery.

WMTS	TMS	WMS-C
1.0.0	1.0.0	1.1.1

GeoServer Natural Earth Vector data

Natural Earth feature data.

WFS	WFS	WFS
2.0.0	1.1.0	1.0.0

GeoServer Natural Earth Raster data and imagery

Direct access to Natural Earth raster data and imagery.

WCS
2.0.1

← Title

← Welcome

← Services

Layer

Layer context:

- Shows Layer Title and Description
- Quick access to common formats!
- Links (with quick filter) for:
 - Metadata
 - Map Formats
 - Feature “Samples” (50 feature default)
 - Tiled Preview
 - Data Links
- Layer virtual services

Show what has been configured for a layer!

The screenshot shows the GeoServer web interface for a layer named 'Countries'. The breadcrumb trail at the top is 'Global > ne > countries'. The page title is 'Countries'. Below the title, it says 'GeoServer Virtual Web Service, anonymous access to ne:countries layer.' and 'Country boundaries on land and offshore. For more information visit [Natural Earth](#).' There is a section 'Preview maps and data' with a 'Filter formats...' input field. Below this are several sections of links: 'Metadata Links' with 'Admin 0 - Countries'; 'Common Formats' with 'OpenLayers', 'GML', and 'GeoJSON'; 'Map Formats' with a dropdown arrow and links for 'AtomPub', 'GIF', 'GeoRSS', 'GeoTiff', 'GeoTiff 8-bits', 'JPEG', 'JPEG-PNG', 'JPEG-PNG8', 'OpenLayers', 'OpenLayers 10', 'OpenLayers 2', and 'OpenLayers 3'; 'Feature Samples' with links for 'CSV', 'GML2', 'GML3.1', 'GML3.2', 'GeoJSON', 'Shapefile', and 'text/csv'; 'Tiled Preview' with links for 'APSTILE / jpeg', 'APSTILE / png', 'CBMTILE / jpeg', and 'CBMTILE / png'; and 'Data Links' with 'ZIP'. At the bottom, there are two sections: 'GeoServer Natural Earth Maps' with links for 'WMS 1.3.0' and 'WMS 1.1.1', and 'GeoServer Natural Earth Tiles' with links for 'WMTS 1.0.0', 'TMS 1.0.0', and 'WMS-C 1.1.1'.

← Layer Title

← Layer Abstract

← Links!

← toggle open

← toggle closed

← Services

Layer Group

Layer group context:

- Layer Group Title and Description
- Quick access to common formats!
- Links (with quick filter) for:
 - Metadata
 - Map Formats
 - Tiled Preview
 - Data Links
- Layer virtual services

The screenshot shows the GeoServer web interface for a layer group named 'World Map'. The breadcrumb navigation at the top reads 'Global > ne > world'. The main heading is 'World Map', followed by a description: 'GeoServer Virtual Web Service, anonymous access to **ne:world** layer group.' Below this is a paragraph: 'World map constructed with Natural Earth countries, boundaries and populated places. For more information visit [Natural Earth](#).' A section titled 'Preview maps and data' contains a 'Filter formats...' input field. Under 'Common Formats', the 'OpenLayers' button is selected. The 'Map Formats' section shows buttons for 'AtomPub', 'GIF', 'GeoRSS', 'GeoTiff', 'GeoTiff 8-bits', 'JPEG', and 'JPEG-PNG'. The 'Tiled Preview' section shows buttons for 'APSTILE / jpeg', 'APSTILE / png', 'CBMTILE / jpeg', and 'CBMTILE / png'. Below these are three sections for services: 'GeoServer Natural Earth Maps' (with WMS 1.3.0 and 1.1.1 buttons), 'GeoServer Natural Earth Tiles' (with WMTS 1.0.0, TMS 1.0.0, and WMS-C 1.1.1 buttons), and 'GeoServer Natural Earth Vector data' (with WFS 2.0.0, 1.1.0, and 1.0.0 buttons).

← Group Title

← Group Abstract

← Links!

← Services

New full screen layer preview

The screenshot displays a web application interface for a map. On the left is a control panel with the following sections:

- WMS version:** A dropdown menu set to "1.1.1".
- Tiling:** A dropdown menu set to "Single tile".
- Antialias:** A dropdown menu set to "Full".
- Format:** A dropdown menu set to "PNG 24bit".
- Styles:** A dropdown menu set to "Default".
- Filter:** A section with a "CQL" dropdown, "Apply" and "Reset" buttons, and a text input field containing "Enter filter expression...".
- Custom Params:** A text input field containing "e.g. viewparams=year:2026&env=color:red".
- ☐ **Enable Debug Mode**

The main area shows a map of Europe with various countries and cities labeled. A "Found 1 Features" popup window is open over Romania, displaying the following table:

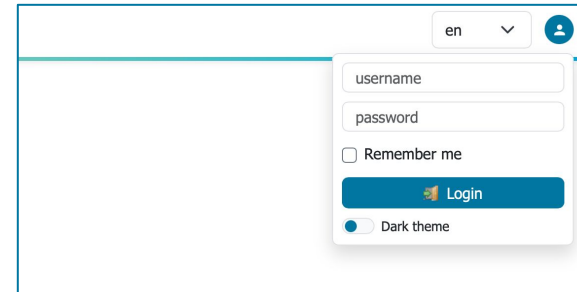
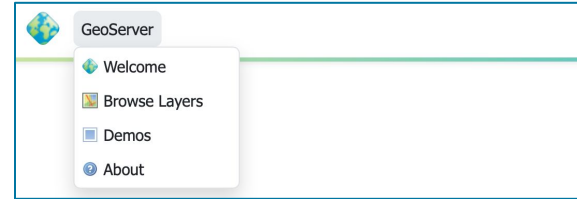
featurecla	Admin-0 countr y
scalerank	1
LABELRANK	3
SOVEREIGNT	Romania
SOV_A3	ROU
ADM0_DIF	0
LEVEL	2

Menu

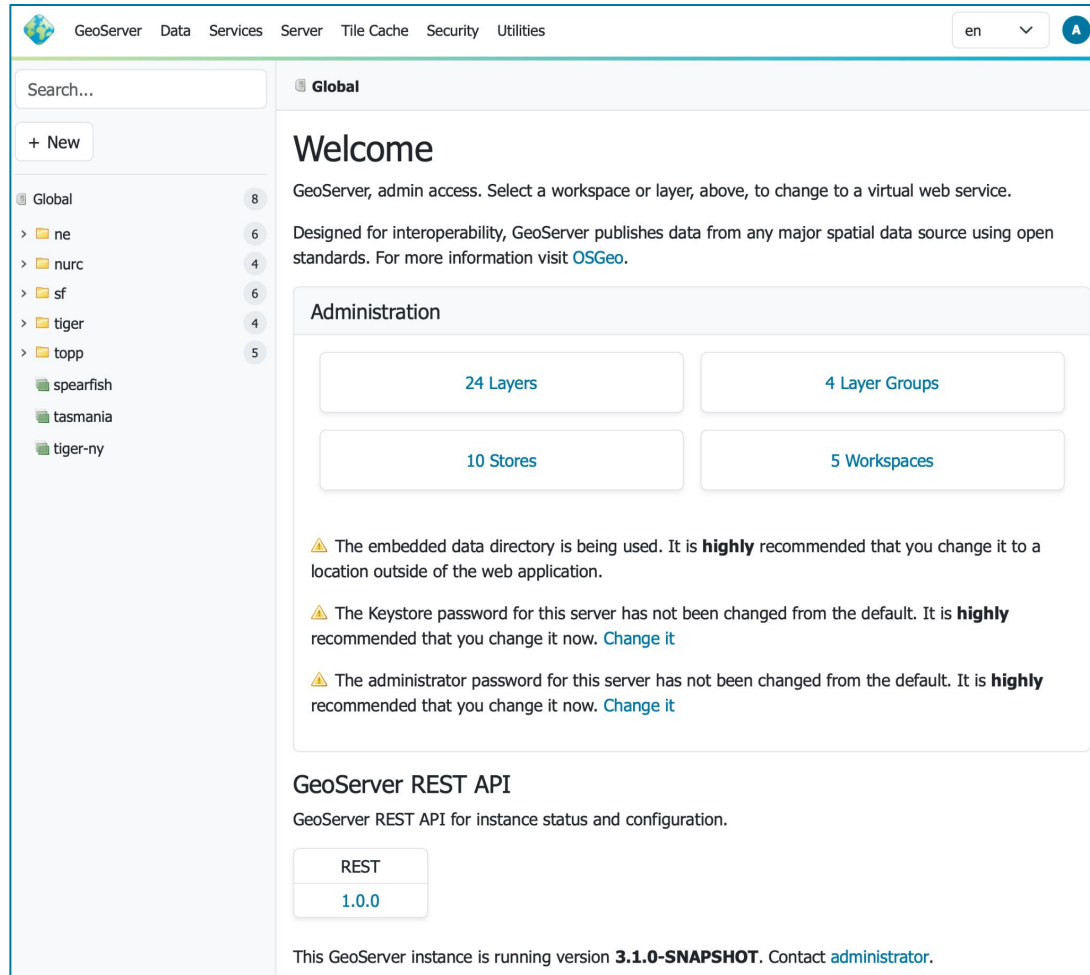
Top-Level menu bar:

- Single “GeoServer” menu for visitors listing pages that do not require login
 - Welcome
 - Browse Layers
 - Demos
 - About
- Language Chooser
- Login

Note: **Layer preview** moved to **Browse Layers** in the GeoServer menu as it does not require a login.



New Admin Experience



The screenshot shows the GeoServer web interface. The top navigation bar includes links for GeoServer, Data, Services, Server, Tile Cache, Security, and Utilities. A language dropdown is set to 'en' and a user profile icon is visible.

Navigation Menu (Left):

- Global (8)
- ne (6)
- nurc (4)
- sf (6)
- tiger (4)
- topp (5)
- spearfish
- tasmania
- tiger-ny

Main Content Area:

Global

Welcome

GeoServer, admin access. Select a workspace or layer, above, to change to a virtual web service.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards. For more information visit [OSGeo](#).

Administration

- 24 Layers
- 4 Layer Groups
- 10 Stores
- 5 Workspaces

Warnings:

- The embedded data directory is being used. It is **highly** recommended that you change it to a location outside of the web application.
- The Keystore password for this server has not been changed from the default. It is **highly** recommended that you change it now. [Change it](#)
- The administrator password for this server has not been changed from the default. It is **highly** recommended that you change it now. [Change it](#)

GeoServer REST API

GeoServer REST API for instance status and configuration.

REST
1.0.0

This GeoServer instance is running version **3.1.0-SNAPSHOT**. Contact [administrator](#).

←Shortcuts

←Feedback

Context Driven Workflow

Global > ne > countries

Countries

GeoServer Virtual World to **ne:countries** layer.

Country boundaries for more information visit [Natural Earth](#).

Administration

[Edit Layer](#) [Edit Store](#)

[Edit Workspace](#)

Preview maps and data

Filter formats...

Metadata Links [Admin 0 - Countries](#)

Common Formats [OpenLayers](#) [GML](#) [GeoJSON](#)

Edit Layer

Edit layer data and publishing

ne:countries

Configure the resource and publishing information for the current layer

[Data](#) [Publishing](#) [Dimensions](#) [Tile Caching](#) [Security](#)

Basic Resource Info

Store Name: NaturalEarth

Native Name: countries

Enabled ☒

Advertised ☒

Name

Title ☐ i18n

Abstract ☐ i18n

WMS version:

1.1.1

Tiling:

Single tile

Antialias:

Full

Format:

PNG 24bit

Styles:

countries

Filter:

CQL

Enter filter expression...

Custom Params:

e.g. viewparams=year:2026&env=color:

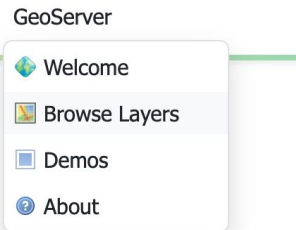
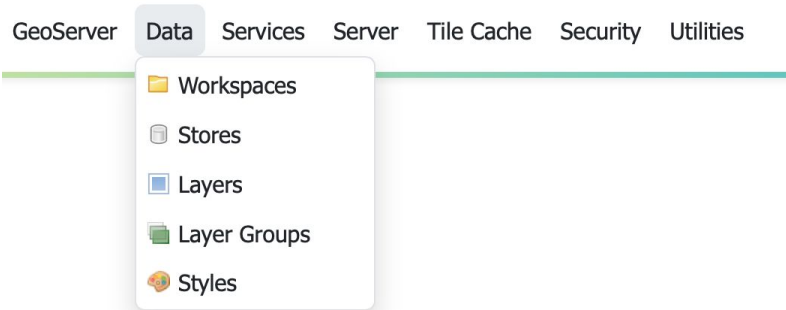
☐ Enable Debug Mode



Data

- Largely unchanged from GeoServer 2:
 - Workspace
 - Stores
 - Layers
 - Layer Groups
 - Styles

Note: **Layer preview** moved to **Browse Layers** in the GeoServer menu as it does not require a login.



Services

Services have been renamed by category to make space for the additional OGC API services.

- Mapping
- Vectors
- Rasters
- Catalog
- Process
- Tiling

Service configuration:

- Divided into Tabs
- Enable/Disable by Version

Reminder:
Layer Editor can also disable Services

GeoServer Data **Services** Server Tile Cache Security Utilities

- Mapping
- Vectors
- Rasters
- Catalog
- Processing
- Tiling

Mapping Services

Mapping, visualization and cartography services publishing visualized geospatial data as map layers.

Workspace ⓘ

Workspace

Service WMS

Service Metadata

Title ⓘ i18n

Maps

Description ⓘ i18n

Georeferenced map data, and imagery.

Vector Services

Services providing discovery and query of vector data as geospatial features.

Workspace ⓘ

Workspace

Service WFS Features

WFS Service Control

Disabled Versions

Available Versions

WFS 1.0.0
WFS 1.1.0
WFS 2.0.0

Vector Services

Services providing discovery and query of vector data as geospatial features.

Workspace ⓘ

Workspace

Service WFS **Features**

Feature Service

Feature Service is a modular OGC API service with "building blocks" that can be enabled and configured for use.

OGC API - Features

Enabled	Name	Identifier	Level	Type
☒	Coordinate Reference Systems by Reference	http://www.opengis.net/spec/ogcapi-features-2/1.0/conf/crs	STANDARD	EXTENSION

Server

Server menu manages the application as a whole, it also provides access to information such as logs that requires login to view.

- **Contact Information** includes a new **Title** setting allowing you to configure the “Welcome” shown to visitors.

GeoServer Data Services **Server** Tile Cache Security Utilities

Contact Information
Server Status
Logs
Global Settings
Image Processing
Raster Access

Contact Information

Set the contact information for this server.

Welcome

Title ☐ i18n

My GeoServer

Description ☐ i18n

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards.

My GeoServer

GeoServer, admin access. Select a workspace or layer, above, to change to a virtual web service.

Designed for interoperability, GeoServer publishes data from any major spatial data source using open standards. For more information visit [OSGeo](#).

Tile Cache

The GeoWebCache powered tile services have been brought together under a single menu.

- Tile Layers
- Tile Cache Settings
- Gridsets
- Disk Quota
- BlobStores

GeoServer Data Services Server **Tile Cache** Security Utilities

- Tile Layers
- Tile Cache Settings
- Gridsets
- Disk Quota
- BlobStores

BlobStores

Configure different BlobStores for the embedded GeoWebCache.

 Add new

 Remove Selected

<< < > >> Results 0 to 0 (out of 0 items)

☐ Identifier BlobStore Type

BlobStore

Configure a BlobStore for the embedded GeoWebCache.

Type of BlobStore: File BlobStore

BlobStore configuration

Identifier * tiles

Enabled ☒

Default ☐

Base Directory * Browse

File System Block Size * 4096

Tiles directory layout GeoWebCache default

Security

Security options have grown for GeoServer 3:

- Settings
- Authentication
- Passwords
- Users, Groups, Roles
- Data
- URL Checks
- Content Security Policy
- CORS - new built-in CORS config
- Services

Previously CORS required editing **web.xml** by hand.

-  Settings
-  Authentication
-  Passwords
-  Users, Groups, Roles
-  Data
-  URL Checks
-  Content Security Policy
-  CORS
-  Services

CORS Settings

Configure settings to control the Cross Origin Request Headers

CORS Settings

Enabled ☐

Allowed Origin Patterns

*

Allowed Methods

GET
POST
HEAD
OPTIONS

Allowed Headers

X-Requested-With
Content-Type
Accept
Origin

Max Age (seconds)

3600

Supports Credentials ☐

Note: GeoServer must be restarted for these changes to take effect.

Utilities

Additional Utilities requiring admin access:

- Extensions like web-resource show up here

GeoServer Tools

Collection of GeoServer administrative tools

[Catalog Bulk Load Tool](#) Tool to bulk copy test data into the catalog. Select the item to copy and how many copies to insert. No need to select a ResourceInfo, selecting only a workspace will create workspace and namespace copies, selecting workspace and store will copy the store.

[Resource Browser](#) Browse and edit resources.

Catalog Bulk Load Tool

Tool to bulk copy test data into the catalog. Select the item to copy and how many copies to insert. No need to select a ResourceInfo, selecting only a workspace will create workspace and namespace copies, selecting workspace and store will copy the store.

Workspace and Namespace

Store

Resource and Layer

of times to duplicate the selected CatalogInfo

Suffix to append to the resource name (plus the count)

Recursive copy ☐

Resource Browser

Browse and edit resources.

Upload Download New resource Edit Cut Copy Paste Rename Delete

- /
- README.rst
- coverages
- csd
- csd.xml
- data
- demo
- global.xml
- gwc
- gwc-gs.xml
- gwc-layers
- layergroups
- layouts
- legendsamples
- logging.xml
- logs
- palettes
- plugins
- security
- styles

Forms

Forms: New two column layout, familiar contents

The image displays two screenshots of the GeoServer web interface, specifically the 'Edit Layer' form for the 'topp:states' layer. The interface is designed with a two-column layout and a sidebar for navigation.

Left Screenshot (Basic Resource Info tab):

- Header:** GeoServer Data Services Server Tile Cache Security Utilities
- Search:** Search in topp...
- Navigation:** + New, Global, topp, states
- Layer List:** Global (8), topp (5), states, tasmania_cities, tasmania_roads, tasmania_state_boundaries, tasmania_water_bodies
- Section: Edit Layer**
 - Sub-section: Edit layer data and publishing
 - Section: topp:states
 - Text: Configure the resource and publishing information for the current layer
 - Tabs: Data, Publishing, Dimensions, Tile Caching, Security
- Section: Basic Resource Info**
 - Store Name: states_shapefile
 - Native Name: states
 - Enabled: ☒
 - Advertised: ☒
 - Name: states
 - Title ☐ i18n: USA Population
 - Abstract ☐ i18n: This is some census data on the

Right Screenshot (Publishing tab):

- Header:** GeoServer Data Services Server Tile Cache Security Utilities
- Search:** Search in topp...
- Navigation:** + New, Global, topp, states
- Layer List:** Global (8), topp (5), states, tasmania_cities, tasmania_roads, tasmania_state_boundaries, tasmania_water_bodies
- Section: Edit Layer**
 - Sub-section: Edit layer data and publishing
 - Section: topp:states
 - Text: Configure the resource and publishing information for the current layer
 - Tabs: Data, Publishing, Dimensions, Tile Caching, Security
- Section: Mapping**
 - Text: Mapping, visualization and cartography services.
 - Section: Cartography Settings
 - Default Style: population
 - Legend Graphic:
 - < 2M
 - 2M - 4M
 - > 4M
 - Boundary
 - Additional Styles:
 - Available Styles: burg, capitals, cite_lakes, dem, generic, giant_polygon, grass, green, line, ne:boundary_lines
 - Selected Styles: pophatch, polygon
 - Opaque: ☐

← Tabs

← Heading

← Section

Updates to tables, buttons and links

- Use of tables
- Buttons → **perform an action**
- Links → **take you to a new screen**

Passwords

Password settings

Active keystore password provider: default

[Change password](#)

Keystore password forgotten?

Change Keystore Password

Change the GeoServer keystore password

Keystore password provider default

Current password:

New password:

Confirmation:

Layers

Manage the layers being published by GeoServer.

[Add a new layer](#) [Remove selected layers](#)

<< < 1 > >> Results 1 to 5 (out of 5 items)

☐	Type	Title	Name	Store	Enabled	Native SRS
☐		Boundary Lines	ne:boundary_lines	NaturalEarth	✓	EPSG:4326
☐		Coastlines	ne:coastlines	NaturalEarth	✓	EPSG:4326
☐		Countries	ne:countries	NaturalEarth	✓	EPSG:4326
☐		Disputed Areas	ne:disputed_areas	NaturalEarth	✓	EPSG:4326
☐		Populated Places	ne:populated_places	NaturalEarth	✓	EPSG:4326

REST API

Yes this link goes to the REST API
(please stop reporting as a bug)

Extents and Keywords

Improvements:

- Previously Keywords were very confusing!
Now a normal table with language selector
- Extent caused some confusion with order.
Now extent shown in compass directions
(but still listed in tab order for data entry)

Layer Keywords Add Keyword

Keyword	Vocabulary
features	(default language) ▼ ✖
countries	(default language) ▼ ✖

Bounding Boxes

Native Bounding Box

Min X

Max X

Min Y

Max Y

83.64513

Max Y

-180

Min X

-90

Min Y

180

Max X

Compute from data

Compute from SRS bounds

Lat/Lon Bounding Box

Min X

Max X

Min Y

Max Y

83.64513

Max Y

-180

Min X

-90

Min Y

180

Max X

Compute from native bounds

GeoServer 3 Upgrade

No Changes to Data Directory

Straightforward upgrade:

- **no changes** to the data directory
- Make it easy!
(to encourage everyone to upgrade)



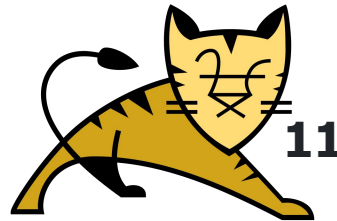
"We've done the upgrade to the 3.x series this month, and it was probably the most boring major upgrade I've done in a long time. It's just chugging along." *Hyperman on Geoserver-Users list*

Minimum requirements

1. JDK 17 minimum (and recommended)
2. Java 21 and 25 tested too (fewer deploys)
3. Tomcat 11 or Jetty 12



More in general, any container supporting Jakarta Servlet 6.1



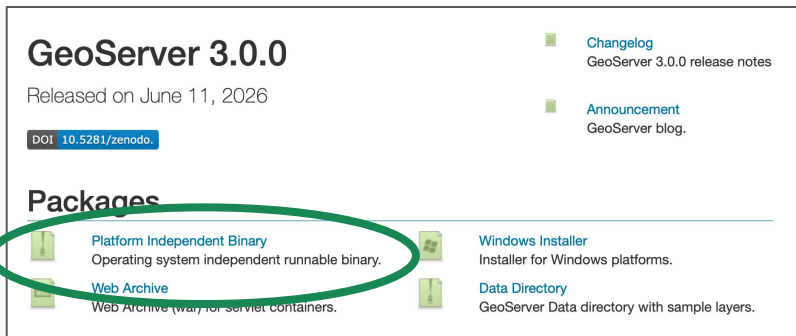
Upgrade Cheat Sheet: GeoServer Binary

General upgrade process:

1. Backup data directory
(copy to an additional place)
2. Make sure data directory is external
(not in application file structure)
3. Write down extensions you have installed
4. Download
 - **Platform Independent Binary**
 - Any **extensions** you have installed
5. Check “upgrade guidance” in docs
6. During startup check logs for warnings

Environment:

- **Java 17** (same as GeoServer 2.28)
- Binary download includes **Jetty 12**



GeoServer 3.0.0

Released on June 11, 2026

DOI [10.5281/zenodo.10.5281/zenodo](https://doi.org/10.5281/zenodo.10.5281/zenodo)

[Changelog](#)
GeoServer 3.0.0 release notes

[Announcement](#)
GeoServer blog.

Packages

Platform Independent Binary Operating system independent runnable binary.	Windows Installer Installer for Windows platforms.
Web Archive Web Archive (may not work for server containers).	Data Directory GeoServer Data directory with sample layers.

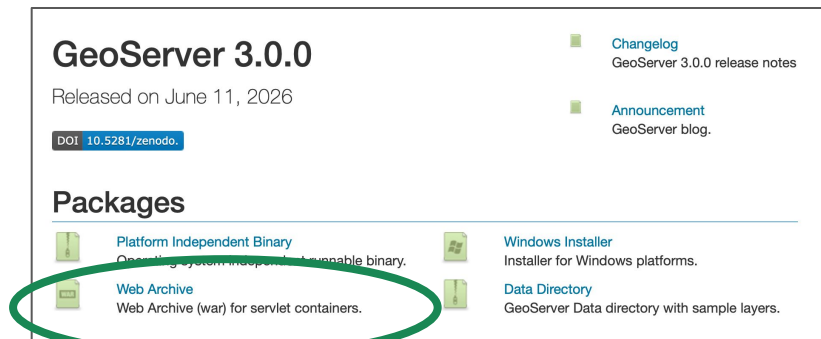
Upgrade Cheat Sheet: GeoServer Web Archive (WAR)

General upgrade process:

1. Backup data directory
(copy to an additional place)
2. Make sure data directory is external
(not in application file structure)
3. Write down extensions you have installed
4. Download
 - **Web Archive**
 - Any **extensions** you have installed
5. Check “upgrade guidance” in docs
6. During startup check logs for warnings

Environment:

- **Java 17** (same as GeoServer 2.28)
- **Tomcat 11.0** is Required



GeoServer 3.0.0

Released on June 11, 2026

DOI 10.5281/zenodo.

[Changelog](#)
GeoServer 3.0.0 release notes

[Announcement](#)
GeoServer blog.

Packages

	Platform Independent Binary Operating system independent, runnable binary.		Windows Installer Installer for Windows platforms.
	Web Archive Web Archive (war) for servlet containers.		Data Directory GeoServer Data directory with sample layers.

Log file location only via sys/env var

The log file location no longer defined in settings (UI or REST API).

Manage using application property:

- Java system property:

```
-DGEOSERVER_LOG_LOCATION=/path/to/geoserver.log
```

- Environmental variable:

```
export GEOSERVER_LOG_LOCATION=/path/to/geoserver.log
```

- Servlet context parameter:

```
<context-param>  
  <param-name>GEOSERVER_LOG_LOCATION</param-name>  
  <param-value>/path/to/geoserver.log</param-value>  
</context-param>
```

Logging Settings

~~Log location~~

~~logs/geoserver.log~~

Logging profile

DEFAULT_LOGGING
GEOSERVER_DEVELOPER_LOGGING
GEOTOOLS_DEVELOPER_LOGGING
PRODUCTION_LOGGING
QUIET_LOGGING
TEST_LOGGING
VERBOSE_LOGGING



Modules downgraded to extension

Some modules downgraded from core to extensions:

- WCS 1.0
- WCS 1.1
- World image support
- ArcGRID support

Ready to install and download if you need them.

OGC Services

- CSW
- CSW ISO Metadata Profile
- WCS 1.0
- WCS 1.1
- WCS 2.0 EO
- WPS
- WPS Clustering (Hazelcast, JDBC)
- WPS download
- WMTS multi-dimensional

Coverage Formats

- ArcGrid
- GDAL
- GRIB
- Image Pyramid
- JPEG2K
- NetCDF
- Raster Attribute Table
- WorldImage

Oauth/OpenID connect

- The previous **OAuth** and **OpenID Connect** module are now retired due to upgrading Spring Security.
- **Rewritten completely** from scratch as a new **supported** module (old modules were community ones)
- The old Keycloak native protocol support is discontinued, but we can use Keycloak OIDC (it's actually tested with it!)
- The new configuration is incompatible.
No panic, we have a detailed [migration guide](#)



H2 database support dropped

H2 embedded database, version 1.x, is completely gone, no way to get it back:

- Why? Continued H2 use annoying!
(project did not maintain on-disk compatibility)

Uses cases removed in the jump between 2.28.x and 3.x:

- **Vector store extension** → move to GeoPKG
- **Disk quota** → use HSQL or, better, a full relational database (e.g., PostgreSQL)



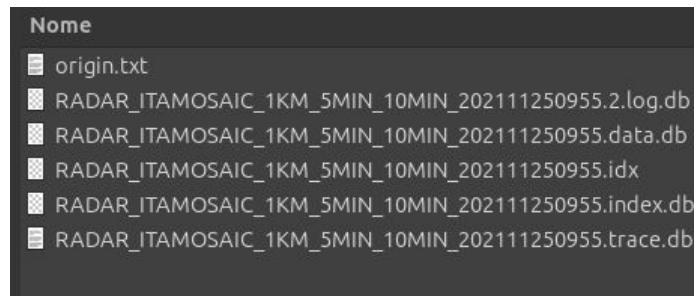
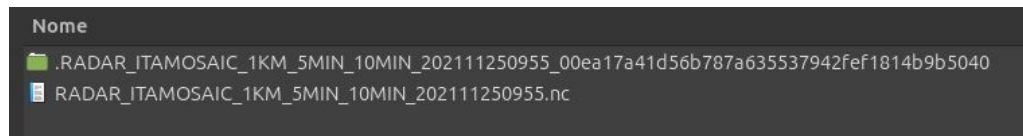
NetCDF/GRIB, no more sidecar dirs

NetCDF indexing simplified:

- No longer needs the sidecar directory
- No longer needs dedicated index database

Consequences:

- Faster config/ingestion
- Same performance
- No changes needed, but you might want to clean up the sidecar files/tables



In summary....

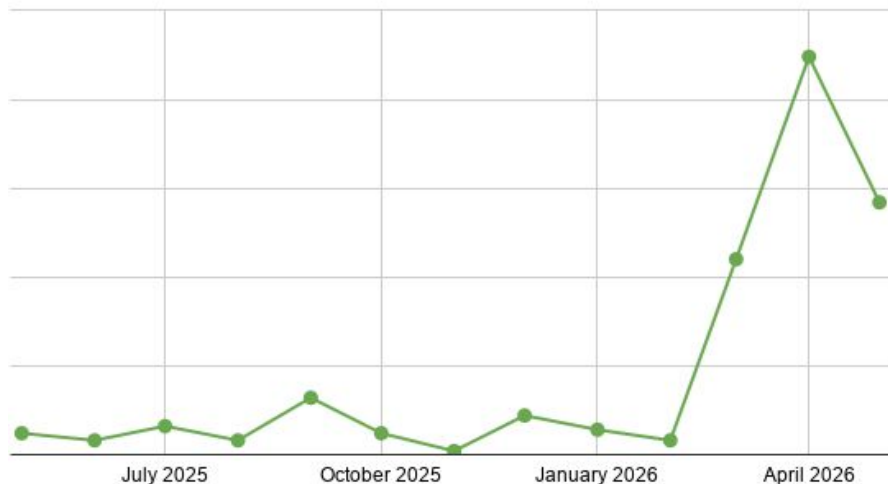
Why? Updates provide a real sense of urgency

The security landscape is under stress:

- Very glad to have completed this project!
 - Regulation is starting to require projects to update dependencies
 - There has been an AI driven increase in the number of security reports
- Successful completion means GeoServer 3 includes the latest Spring Framework 7

[Spring and Security In The Times Of AI](#)

Community Security Reports for Spring By Month



GeoServer 3 Tour Recap

Straightforward Upgrade

1. Data directory unchanged
2. Java 17
3. Tomcat 11 (or Jetty 12) Required
4. Core is lighter! Use extensions for:
WCS 1.0, WCS 1.1, World image, ArcGRID
5. OAuth OpenID Connect (OIDC) Extension
Migration from OAuth2 and KeyCloak
6. H2 no longer used for local database
7. NetCDF indexing now self contained
(Can clean up some unused files)

Context-Driven User experience:

- Quick search
- Context shown as breadcrumbs
- Page content adapts follow context
- Context menu for common actions
- Menu bar for global navigation menus

GeoServer Theme:

- Clean look, Responsive, light/dark mode

A round of applause to our sponsors please!



Individual donations: Abhijit Gujar, Hennessy Becerra, Ivana Ivanova, John Bryant, Jason Horning, Peter Smythe, Sajjadul Islam, Sebastiano Meier, Stefan Overkamp.



Thank you!



Default

Primary

Success

Info

Warning

Primary

Success

Info

Warning

```
/* In CSS */  
color: var(--gs-primary);  
background: var(--gs-danger-bg-subtle);  
border-color: var(--gs-border-color);
```



GeoServer